

Quantitative Estimation

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SurvMeth632

Outline

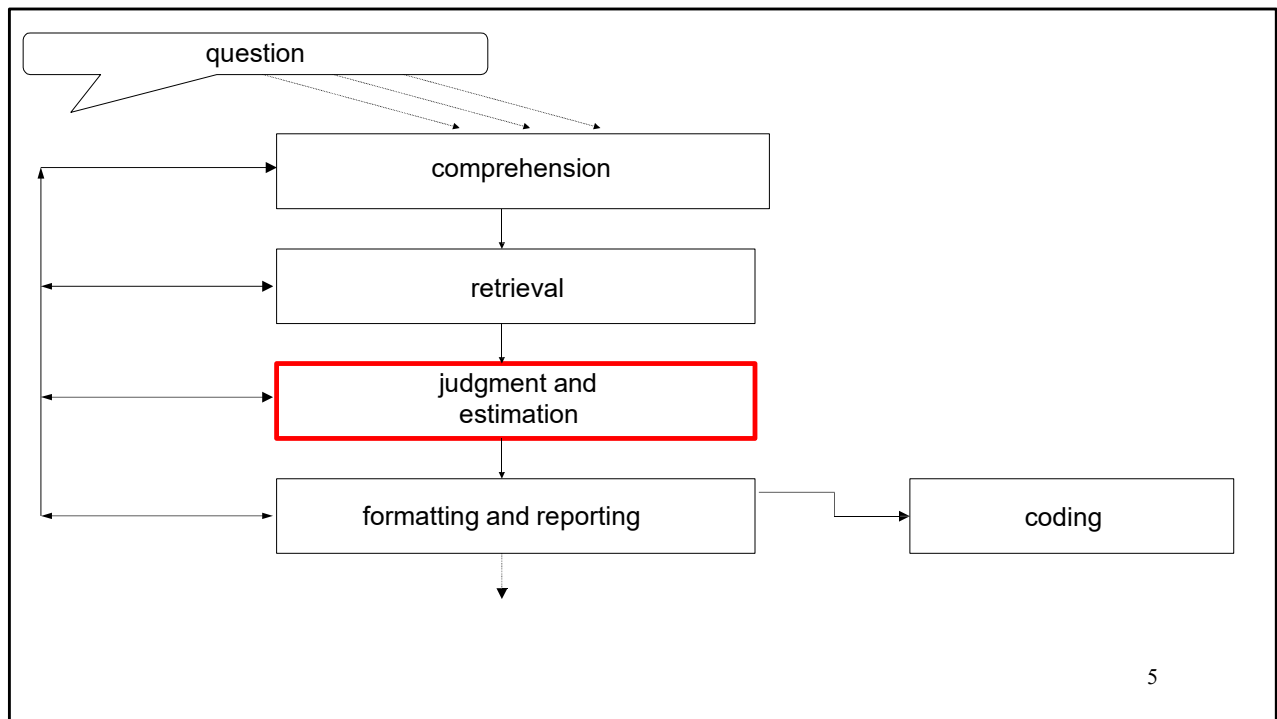
- Metrics and Mapping Framework
 - Deriving Metric Information, e.g., from response scales
 - Deriving Mapping Information, e.g., automatic encoding of frequency
 - Meta-Memory strategies: Availability Heuristic
 - Other heuristics: Representativeness, Anchor and Adjust
- Decomposition Strategies
 - Improved recall versus overestimation
- Strategies for Estimating Behavioral Frequency
 - Multiple Strategy Perspective: Enumeration, Rates, and Non-numerical strategies
 - Estimating emotion frequency

Compensating for Imperfect Recall

- Recall rarely provides veridical record of events
 - memory degrades as elapsed time increases
 - similar episodes may merge into generic memories
 - retrieval cues may not bring entire event or all events to mind
- When recall cannot serve as basis of numerical response, respondents may estimate
 - Can be a conscious choice
 - Respondent may recognize limits of recall
 - Respondent may believe task does not require recall
 - May be the only option
 - recall does not provide sufficient information to perform task

Compensating for Imperfect Recall (2)

- Estimation requires some “raw material”
 - may come from retrieval stage in response framework
 - or retrieval stage may be skipped entirely, and raw material found elsewhere: e.g., response scale

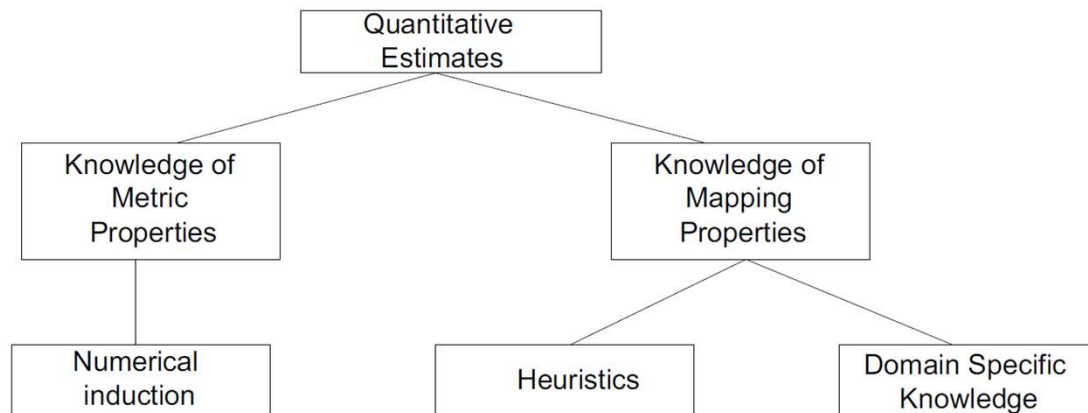


Metrics and Mapping Framework

(Brown & Siegler, 1993)

- Quantitative estimation involves:
 1. *metric knowledge*: statistical properties of domain, such as mean, variance and form of distribution
 - e.g., there are four time zones in the continental US so if noon here, 9AM in CA
 2. *mapping knowledge*: ordinal relations (position) of entities on relevant dimension
 - derived through heuristics and domain knowledge
 - Example of heuristic: availability (Tversky & Kahneman) provides a sense of relative magnitude
 - Example of domain knowledge: knowing when Joe Biden became president (2021) may help a person establish when COVID vaccines became available

Metrics and Mappings Framework, Basic Structure



Deriving Metric Information

- If respondents can retrieve numerical facts, can use these as basis of estimate
 - e.g., I donate blood every eight weeks so donated 6 times last year
- If respondents lack info about the numerical properties of possible answers, may use info in question, such as response scale
- Schwarz, Hippler, Deutsch & Strack (1985) observed use of response scale values to infer distributional properties of domain (in their case, hours of TV watching)

Scales in Schwarz et al. (1985)

Low frequency alternatives	%	High frequency alternatives	%
Up to $\frac{1}{2}$		Up to $2\frac{1}{2}$	
$\frac{1}{2}$ to 1		$2\frac{1}{2}$ to 3	
1 to $1\frac{1}{2}$		3 to $3\frac{1}{2}$	
$1\frac{1}{2}$ to 2		$3\frac{1}{2}$ to 4	
2 to $2\frac{1}{2}$		4 to $4\frac{1}{2}$	
> $2\frac{1}{2}$		> $4\frac{1}{2}$	

- On Low Frequency scale, 16.2% report > 2.5 hrs. of weekly TV watching; on High Frequency scale, 37.5% report > 2.5 hrs. of weekly TV watching
- Low Freq respondents rate mean for average citizen lower (2.7 hrs.) than High Freq respondents (3.2 hrs)

Deriving Metric Information (2)

- Respondents take response alternatives into account to a greater extent the less they know about the actual quantities involved
 - proxy respondents' reports are more affected by scale values than are respondents' reports about themselves (Schwarz & Bienias, 1990)
 - reports about irregular behaviors more affected by scale values than reports about regular behavior (Menon, Raghurir & Schwarz, 1995)
 - Chinese respondents unaffected by scale values for observable events (e.g., being late for class) but are affected by scale values for private events (e.g., telling a lie) (Ji, Schwarz & Nisbett, 2000)
 - when respondents allowed to review last week's TV listings, effect of scale values goes away (Chasien, Strack & Schwarz, 1986)

Mapping Info: Automatic Encoding of Relative Frequency

- Relative information about quantities (e.g., smaller versus bigger) may be encoded without awareness
 - Should enable people to estimate frequencies without counting
- Lab study (Alba, et al., 1980):
 - Presented series of words, one at a time, which participants were asked to read aloud
 - Words belonged to different categories: birds, trees, fruit, etc.
 - Surprise test on frequency (# instances) in each category
 - e.g., how many birds?
 - Participants given different response deadlines so that, if recall-and-count possible, would expect more of it when given more time
 - Recall-and-count requires time to recall each instance

Automatic encoding of relative frequency (2)

Actual Frequency	Test rate (deadline in secs.)				Mean Judgment
	2	4	5	10	
0	0.55	0.35	0.40	0.10	0.35
3	3.37	2.93	2.62	2.60	2.88
6	4.62	3.85	4.33	4.24	4.26
9	5.91	5.08	5.20	5.58	5.44
<i>M</i>	3.61	3.05	3.14	3.13	

Judged Frequency

Alba et al. (1980)

- Alba et al. (cont'd): excellent relative accuracy ("Mean Judgment");
- no effect deadline condition
 - suggests this is not based on recall of explicitly encoded frequencies
- Automatically encoded relative information
 - may be backup that most survey respondents have available

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Deriving Mapping Information: Heuristics

- Procedures that provide “good enough,” but not optimal, solutions
 - mental shortcuts for quickly solving problems in a way that is useful, most of the time,
 - when time is limited,
 - effort is lacking
 - Can be more accurate than algorithm when computation hard
 - Often produce relative information
 - e.g., $x > y$ rather than absolute value, e.g., $x=12$

Should I ride my bike or take the bus to work?

1. Heuristic:

- when lots of traffic, faster to ride my bike
- it's rush hour now so I will ride my bike

2. Algorithm:

- it takes between ~8 and ~21 minutes to ride the bus to work depending on traffic conditions
- it takes ~15 minutes to ride my bike to work irrespective of traffic
- according to my transit app, the bus will arrive in 10 minutes so will take 18 minutes, i.e., > 15 minutes
- I will ride my bike

Heuristics: Metamemory strategies

- When estimating quantities, people may consider their beliefs about how human memory works
 - If it is hard to recall info about a topic or event, one might infer that is because the event is rare
 - *Lack of knowledge inference* (Gentner & Collins, 1981): If one can't remember something that should be memorable, one might infer it did not happen

Availability Heuristic

- Tversky & Kahneman, 1974 (original idea):
 - people sometimes use the ease with which examples come to mind as an indication of their frequency or probability of occurrence
 - famous names easier to recall than less famous names; when list had more famous females than males, females judged to be more frequent even though equal numbers males and females in list
- Schwarz, et al., 1991 (application to survey-like task):
 - Asked respondents to list either 6 or 12 situations in which they were assertive
 - Then asked to rate own assertiveness
 - Respondents who listed 6, rated themselves as more assertive than respondents who listed 12
 - Presumably, easier to bring fewer instances to mind so respondents asked to list 6 situations in which they were assertive conclude more likely to be assertive

Other Heuristics: Representativeness

- Tversky & Kahneman, 1974: estimates based on similarity to target rather than on *base rates*
 - descriptions of people that *sound like* engineers judged more likely to be engineers despite the odds (.7/.3 lawyer)
 - “Jack is a 45-year-old man. He is married and has 4 children. He is careful and ambitious. He shows no interest in political or social issues and spends much of his free time on his hobbies which include home carpentry, sailing and mathematical puzzles”
 - Effect persists even when descriptions are neutral: people rate chances of being an engineer as .5 despite base rate
 - T&K got people to use base rates when that was all they gave them

Other Heuristics: Anchor and Adjust

- Anchor and adjust (Tversky & Kahneman, 1974):
 - answers to numerical questions may involve partial computation and extrapolation
 - former serves as anchor, latter modifies (adjusts) before reporting
 - Participants asked to multiply series of numbers; were interrupted before could complete
 - series ran from highest to lowest ($8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$) or vice versa
 - Interrupting participants who start at high end produces higher partial product (anchor) than interrupting participants who start at low end, leading to higher anchor
 - High to low led to higher estimates (2250 versus 512), suggesting larger anchor
 - although both series led to severe underestimation (actual value = 40,320), suggesting adjustment is usually inadequate

Decomposition

- Recall may be difficult because categories about which respondent is asked to recall information is large and complex
- Some survey researchers break single category into smaller categories and ask several questions
 - *Decomposition*, e.g., instead of asking about “restaurants” ask about fast food then Chinese then Mexican then pizza then all other restaurants
 - reports for all subcategories summed to produce overall report
- Generally found to increase reported quantities relative to “undecomposed” reports (e.g., total number of restaurant meals)
- Is more necessarily better?

Decomposition (2)

- Belli, Schwarz, Singer & Talarico (2000) tested possibility that frequency of decomposed categories might be overestimated because of availability heuristic
 - asked how many telephone calls made in previous week or 6 months and compared to billing records
 - In Decomposed condition, respondents asked about (a) number of monthly calls over 6-month period and (b) number of local vs. not local calls
 - Compared to Undecomposed (global) reports (i.e., all months and all locations)

Decomposition (3)

- Decomposition led to overreporting
 - For local calls in 6-month period (+136.6)
 - For calls outside local region in 6-month period (+98.6)
 - “Undecomposed” questions led to non-significant underreporting for the same 6-month period, i.e., reasonably accurate
- Availability heuristic as explanation for overreporting
 - easier to recall instances for small than large categories so within each month or location seems like more calls than there are (true value) – leads to overestimation
 - when summed, greater than undecomposed estimates
 - Suggests, at least for non-distinctive events like phone calls, decomposition is to be discouraged

Strategies for answering Behavioral Frequency (BF) questions

- Direct retrieval of a specific frequency is rare
 - e.g., unlikely respondent can directly retrieve exact number of restaurant meals in 2022
- Instead, one approach, proposed by Sudman & Bradburn (1974), is to recall and count, i.e, *enumerate*, events
- Researchers realized that enumeration is supplemented by retrieving rate of occurrence
 - Blair & Burton (1987) argued that, in addition to enumeration, respondents also use non-enumeration strategies, primarily rate of occurrence, e.g., “2 times a week”
 - should lead to different patterns of error than enumeration
 - should be used more when enumeration is overwhelmed by number of episodes
 - enumeration and rates seen as complementary -- one or the other

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Rates vs. Enumeration

- To assess strategy use, Blair and Burton (1987) asked respondents to *think aloud* immediately after answering six BF questions
- Classified response strategy based on think aloud:
 - 28% enumeration, 56% rates, and 9% “other”
- Enumeration should be less common and use of rates more common when
 - reported frequency is relatively large
 - reference period is relatively long
 - question worded “How often” versus “How many times”

Rates vs. Enumeration (2)

- Enumeration less common, rates more common when
 - 1) reported frequency relatively large:
 - 84% of responses ≤ 3 based on enumeration
 - 0% of responses > 10 based on enumeration
 - 2) reference period relatively long:

Reference period	% Enumeration	% Rates
2 weeks	56	36
2 months	25	54
6 months	4	76

- Respondents' realization that recall task too hard is a type of metamemory (or metacognition)

Rates vs. Enumeration (3)

- Enumeration less common when question is “How often?” than “How many times?”
 - when asked “How often” in 6 questions, respondents answered as a rate (e.g., “once a day”) more than absolute number (e.g. 7)
- Unanswered questions:
 - How does strategy affect accuracy?
 - Are there other strategies beside enumeration and rate-based responding?
 - If so, what are their conditions of use and implications for accuracy?

Strategy and accuracy

- Burton & Blair (1991) examined relationship between strategy and response accuracy using bank and registrar records as gold standard
 - Strategy identified through written verbal reports immediately after responding
- Overall, enumeration led to underestimation and use of rates to overestimation
 - Enumeration more likely and respondents more accurate when given more time to respond (# non-business courses and # courses in which received “B”)
- Burton & Blair interpret the underestimation associated with enumeration as the result of forgetting and the overestimation due to rate-based estimation as the failure to consider exceptions to the routine

Regularity and similarity (Menon, 1993)

- People are most likely to base response on rates when
 1. events occur on *regular* schedule (ensuring that rate info is available)
 2. individual occurrences are *similar* to each other so recalling individual events is difficult
- People most likely to enumerate when
 1. events occur on *irregular* schedule (no rate info available)
 2. individual occurrences are *distinct* from one another (more memorable)
- Strategy and Response Accuracy:
 - When use of rates is most likely, accuracy is highest (assuming behavior does not deviate from regular schedule)
 - When enumeration most likely, accuracy is lowest (forgetting)

Regularity and similarity (2)

- *Rs* maintained diaries for 1 week and then judged frequency of same mundane activities
 - Diaries used as gold standard in calculating response accuracy
- Accuracy was highest for similar-regular behaviors, i.e., when rates most likely to be available and recall is hardest
- Accuracy lowest for dissimilar-irregular events, i.e., when enumeration likely and rates unlikely (not available)
 - No overestimation with rates as was observed by Burton & Blair

	Similar	Dissimilar
Regular	.93	.59
Irregular	.85	.24

Correlation between reported and actual frequency

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Multiple strategy perspective

- Non-numerical strategies available to respondents, in addition to enumeration and rates (which are numerical)
- Particularly likely when
 1. enumeration is difficult, e.g., when actual frequency is high
 2. no rate information is available, e.g., events occur on irregular schedule
 - Non-numerical strategies give *Rs* a relative sense of frequency (mapping info.) but not actual numbers (metric info.)
- These strategies may provide a qualitative sense of magnitude
 - “that happens a lot,” “that hardly ever happens”
- To produce a numerical answer, one must convert relative quantity (mapping info) to number (metric info.)
 - Likely to produce *rounded* estimates

Multiple Strategy Perspective (2)

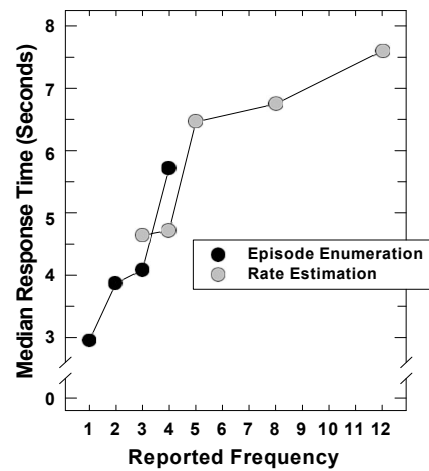
Response Strategy	%	Regularity	Similarity	Mean Reported Frequency	RT Slope
1. Enumeration	27	2.39	2.86	2.3	.84
2. Rate Retrieval	15	3.66	3.54	7.8	.02
3. Rate Estimation	12	3.43	3.31	11.1	.15
4. Rate and Adjust	9	3.65	3.43	11.9	.07
5. General Impression	18	2.91	3.11	12.3	.01
6. Uncodable	18				

Conrad, Brown & Cashman (1998)

- Enumeration used most to report low frequency responses
- Rates used more when events rated more regular and similar
- General Impressions (non-numerical) relatively common (18%); used when reports are largest
 - Brown (1996) showed that general impression strategy leads to overestimation: map qualitative sense of frequency to number line so cannot be smaller than zero but can be very large

Multiple Strategy Perspective (3)

- Responses times (RTs) increase with reported frequency when answers based on Enumeration (including Rate Estimation)



Enumeration and Memory Ability

- Ziniel (2008) examined how reduced memory ability in aging affects selection of estimation strategy
- Reanalyzed Conrad, Brown and Cashman (1998) based on respondent age and observed less enumeration among older respondents

	Young	Old
Episodic Enumeration	36.05%	26.32%
Rate retrieval	17.53%	21.80%
Rate estimation	14.32%	13.53%
Rate and adjustment	12.10%	8.27%
General impression	20.00%	30.08%
	100%	100%
N	405	133

Estimating Emotion Frequency

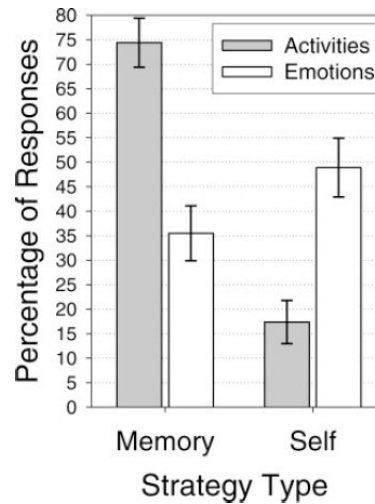
- Mental health questionnaires routinely ask people to estimate how often they experience certain emotions
 - e.g., “In the last 14 day on how many days did you feel nervous?”
- Are people accurate when estimating frequency of emotions?
- Do they use the same strategies they use to estimate frequency of behavior?

Estimating Emotion Frequency (2)

- Brown, Williams, Galambos & Barker (2007) compared accuracy and strategy for estimating frequency of emotion and activity (i.e., behavior)
- *Rs* completed online diary every day for 14 days
 - Responded to emotion and activity check-list
 - Emotions (y/n): energetic, excited, happy, irritable, sad, nervous, upset
 - Activities: # alcoholic drinks, # minutes exercised, socialize with friends (y/n), written quiz (y/n)
- On day 15, respondents estimated number of days out of past 14 in which they experienced or engaged in each emotion and activity
- Then indicated strategy by selecting from menu:
 - Authors recoded 6 categories into *Memory-based* (i.e., remembered events) versus *Self-Knowledge* (i.e., facts about self such as routines)

Estimating Emotion Frequency (3)

- Absolute error (reported frequency versus diary) small for both emotions and activities
- Signed error is small and negative (underestimates)
- Memory-based strategies used more for activities than emotions
- Self-knowledge used more for estimating frequency of emotions than activities



Summary

- Recall not always sufficient for accurate survey responding
- To supplement recalled information, people derive:
 - Quantitative (metric, absolute) information
 - Qualitative (mapping, relative) information
- Heuristics and metamemory strategies (e.g., availability heuristic) produce mapping information that is often “good enough” but sometimes leads to systematic measurement error
- Decomposition of non-distinctive behaviors can increase recall-based strategies but may lead to overestimation by triggering availability heuristic

Summary (2)

- Respondents seem sensitive to memory demands of task (metamemory)
 - They don't enumerate (recall and count) when frequency is high
 - Older respondents (whose memory is weaker than that of younger respondents) don't enumerate as often as use general impressions to estimate frequency
- Multiple strategies available to respondents
 - Each produces signature pattern of error (under- vs. overestimation)
 - Characteristics of events (e.g., regularity, similarity, frequency) affect strategy use
- Estimates of emotion frequency as accurate as estimates of activities
 - based more on self knowledge than enumeration; opposite for activities³⁷